

E-COMMERCE SALES ANALYSIS

CodSoft Internship — Task 2

RECORDS	TOP REVENUE CATEGORY	TOP REGION	South
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A concise business-analysis report based on the Python/Pandas exploration completed for CodSoft Task 2.

Analysis focus: descriptive statistics • distributions • business trends • revenue performance • outlier checks • relationships

1. Objective

The purpose of this analysis is to examine an e-commerce sales dataset using Python and Pandas, understand distributions and trends, identify unusual values, and use summary statistics to answer practical business questions.

Business questions explored

- Which product category receives the most orders?
- Which category generates the highest total and average revenue?
- Which region contributes the most revenue?
- Is there a relationship between quantity sold and revenue?
- Which values appear unusual and may require investigation?

2. Dataset Overview

The dataset contains 1,000 e-commerce orders recorded across 2025. It includes transaction, customer, product, pricing, payment, and rating information.

Field	Description
Order_ID	Unique order identifier
Order_Date	Date of order
Category	Product category
Product	Product purchased
Region	Customer/order region
Customer_Type	New or Returning customer
Quantity	Units purchased
Unit_Price	Price per unit
Discount_Percent	Discount applied
Revenue	Revenue generated
Payment_Method	Payment method
Customer_Rating	Rating from 1 to 5

3. Analysis Approach

- Structure inspection: shape, columns, and data types were examined.
- Descriptive statistics: count, mean, standard deviation, minimum, and maximum were reviewed.
- Grouped analysis: order counts and revenue were compared by category and region.
- Outlier review: Quantity and Revenue were checked for unusually high observations.
- Relationship analysis: correlation was used to examine Quantity-Revenue and Discount-Revenue relationships.

4. Key Findings

4.1 Orders by Category

Fashion recorded the highest number of orders with 223. Sports followed with 209, Beauty with 197, Electronics with 192, and Home with 179. Order frequency and revenue performance are therefore not the same thing.

Category	Orders
Fashion	223
Sports	209
Beauty	197
Electronics	192
Home	179

4.2 Revenue by Category

Electronics generated the highest total revenue at approximately ₹21.82 million, followed by Home at ₹5.98 million. Sports generated ₹3.48 million, Fashion ₹3.04 million, and Beauty ₹1.77 million.

Category	Total revenue
Electronics	₹21,821,749.75
Home	₹5,980,307.15
Sports	₹3,482,830.39
Fashion	₹3,037,052.61
Beauty	₹1,773,920.52

4.3 Average Revenue per Order

Electronics had the highest average revenue per order at approximately ₹113,655. Home averaged about ₹33,410, while Sports, Fashion, and Beauty averaged approximately ₹16,664, ₹13,619, and ₹9,005 respectively. This highlights the importance of product value and order mix.

4.4 Revenue by Region

The South region generated the highest revenue at approximately ₹10.55 million. West followed at ₹9.41 million, North at ₹8.69 million, and East at ₹7.45 million.

5. Outlier & Relationship Analysis

5.1 Outlier Review

Quantity and Revenue were reviewed using descriptive statistics. The dataset contains intentionally unusual high values, including large order quantities and exceptionally high revenue observations. These should be treated as potential outliers rather than automatically deleted; in a real business dataset, each unusual record should be validated against the source system.

5.2 Quantity and Revenue Correlation

The Pearson correlation between Quantity and Revenue is 0.2093, indicating a weak positive relationship. Higher quantities tend to be associated with somewhat higher revenue, but quantity alone does not strongly explain revenue. Product price and transaction mix are likely important factors.

5.3 Discount and Revenue

Discount versus Revenue was examined through correlation in the Python analysis. This check helps assess whether promotional pricing moves together with revenue. Correlation should be interpreted as association, not proof of causation.

6. Business Interpretation

- Electronics is the key revenue driver. Its total and average revenue are substantially above the other categories.
- Fashion leads in order volume. High order frequency does not automatically translate into the highest revenue.
- South is the strongest region by revenue. Regional performance can support sales planning and targeting.
- Quantity is not the only revenue driver. The weak positive correlation suggests product value and pricing matter.
- Outliers need validation. Very large transactions may be genuine and should not be removed without checking.

7. Recommendations

- Investigate the Electronics products driving high revenue per order.
- Study Fashion customers and products to understand how high order volume can be converted into higher order value.
- Review successful South-region sales patterns and assess whether they can be applied elsewhere.
- Segment revenue by product, discount, customer type, and region for deeper decisions.
- Validate extreme Quantity and Revenue records before treating them as data-quality errors.

8. Conclusion

The analysis shows a clear difference between sales volume and revenue performance. Fashion generated the most orders, while Electronics generated the most revenue and the highest average revenue per order. South was the strongest regional contributor. The weak Quantity–Revenue correlation indicates that quantity alone is not enough to explain revenue; product pricing and transaction mix are important. Overall, the project demonstrates how descriptive statistics, grouping,

outlier checks, and correlation can turn raw sales data into practical business insights.

Portfolio note: This report accompanies the Python/Pandas Task 2 project and can be submitted with the CSV dataset and task2_analysis.py script.